

Sharif University of Technology
Department of Mechanical Engineering

Title:

Modelling and Control of a Separately Excited DC Motor in MATLAB/Simulink

Course Project, Fundamentals of Electrical Engineering II

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Abstract. In this project, a separately excited DC motor under a constant-torque load was modelled, simulated and controlled. The main aim of the project is to study the dynamic behaviour of the DC motor, to select a suitable configuration based on the load type, to design a speed controller, and to analyse the simulation results in MATLAB/Simulink.

First, among the loads specified in the assignment, the constant-torque load of $T_L = 1.5\,\mathrm{N}\cdot\mathrm{m}$ was selected. Then, among the four types of DC motor, the separately excited DC motor was chosen. The reason for this choice is the ability to control the field circuit and the armature circuit independently. In this type of motor, the field current can be kept approximately constant, and the speed can be controlled by varying the armature voltage. This feature suits a constant-torque load, because with a constant field flux the motor torque depends mainly on the armature current.

In the first stage, the open-loop model of the motor was built in Simulink. The results showed that with an armature voltage of 24 V, a field voltage of 12 V and a constant load of $1.5\,\mathrm{N}\cdot\mathrm{m}$, the steady-state speed of the motor reaches approximately 697.7 rad/s. Next, a PI controller was designed for speed control. The reference speed was set to 600 rad/s, because this value is lower than the speed achievable by the motor in open loop, so the motor can reach it without exceeding the armature-voltage limit.

The final controller was designed with gains $K_p = 0.035$, $K_i = 0.015$ and $K_d = 0$. In addition, the controller output was limited to the range 0–24 V with a Saturation block. The closed-loop results showed that the PI controller brought the motor speed to 599.4 rad/s, i.e. very close to the reference of 600 rad/s. The steady-state controller output was about 20.7 V, which also agrees with the theoretical calculations.

Keywords: separately excited DC motor, constant-torque load, MATLAB/Simulink, transfer function, torque–speed characteristic, PI speed control, actuator saturation

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1 Introduction

Direct-current (DC) motors are among the most widely used electrical machines in industry. Owing to their good controllability of speed and torque, favourable dynamic response, relatively simple structure and ease of mathematical analysis, they are used in many industrial and research applications, including electric transportation systems, robotics, machine tools, conveyor belts, laboratory equipment and industrial drives. Despite the advances in AC motors and modern drives, DC motors remain one of the main choices for teaching the concepts of control and of modelling electromechanical systems.

One of the most important features of DC motors is the possibility of controlling speed and torque by varying the armature voltage or the field flux. This feature has made DC motors a reference system in many control problems. Moreover, the mathematical relations governing their operation are relatively simple, which makes it possible to derive accurate models for analysing their transient and steady-state behaviour.

Among the different types of DC motor, the separately excited DC motor is of particular importance. In this type of motor, the field circuit and the armature circuit are supplied from independent sources. This independence keeps the magnetic flux of the motor approximately constant, so that speed control is carried out mainly by adjusting the armature voltage. Such a structure offers good flexibility for the design of control systems.

In many industrial applications, electric motors are subjected to different loads, each of which imposes a different behaviour on the system. The load type has a direct influence on the choice of motor and control method. In this project, the selected load is a constant-torque load, as found in equipment such as conveyor belts, hoists and some industrial machinery. Under these conditions, the motor must supply the torque required by the load and maintain the desired speed.

The main aim of this project is the mathematical modelling of a separately excited DC motor, the selection of a suitable configuration for a constant-torque load, the simulation of the dynamic behaviour of the motor in MATLAB/Simulink, and the design of a speed controller. To this end, the electrical and mechanical equations of the motor are first derived and its dynamic model is developed. The performance of the motor is then examined in open loop, after which a speed controller is designed and the performance of the system is evaluated in closed loop.

In the rest of the report, the project problem and the specifications of the selected load and motor are first introduced. The reasons for choosing the motor and load, and the best motor configuration for the load type, are then presented. The following sections cover the mathematical model of the motor, its Simulink implementation, the motor characteristic curve, the open- and closed-loop simulation results, the validation of the results, and the analysis of the controller performance. Finally, the results are summarised and suggestions for improving the performance of the system are given.

2 Problem Statement

The aim of the project is to select a load type and a DC motor type, to build a suitable model of the motor for the selected load, to design a controller for the motor, and to examine the simulation results. The following choices were made in this project:

- load type: constant-torque load;
- load value: $T_L = 1.5\,\mathrm{N}\cdot\mathrm{m}$;

- motor type: separately excited DC motor;
- speed-control target: $\omega_{\text{ref}} = 600\,\text{rad/s}$.

In open loop, the armature voltage is constant and the motor reaches a natural speed determined by its parameters and the applied load. In closed loop, however, the aim is to regulate the motor speed at the desired value of 600 rad/s by means of a controller.

3 Reasons for the Choice of Load and Motor

3.1 Reasons for Choosing the Constant-Torque Load

The load selected in this project is a constant-torque load with the value

$$T_L = 1.5\,\text{N}\cdot\text{m}. \tag{3.1}$$

A constant-torque load is simpler to model than a constant-power load, because its torque is assumed to be independent of speed. In a constant-power load, the relation between torque and speed is inverse, which makes control of the system more complicated.

On the other hand, the value of $1.5\,\text{N}\cdot\text{m}$ is more demanding than very small loads such as $0.01\,\text{N}\cdot\text{m}$ and $0.001\,\text{N}\cdot\text{m}$ and better exhibits the performance of the motor. This load is suitable for studying the transient response of the current, torque and speed, and it makes the simulation results more valuable from an educational and analytical point of view.

3.2 Reasons for Choosing the Separately Excited DC Motor

Among the four available types of DC motor, the separately excited DC motor was selected. The main reason for this choice is its suitability for the selected load and its good controllability.

In a separately excited DC motor, the field circuit and the armature circuit are supplied from two independent sources. The field voltage can therefore be kept constant while the motor speed is controlled by varying the armature voltage. This structure suits constant-torque loads, because with a constant field current the motor flux remains approximately constant and the electromagnetic torque becomes mainly a function of the armature current.

In terms of magnitude, $1.5\,\text{N}\cdot\text{m}$ is neither a very small, light load nor a very heavy one. Using a series motor, which is suited to high starting torque and heavy loads, is therefore not an optimal choice. The permanent-magnet motor is simple but offers less control flexibility. The shunt motor is acceptable but gives less control freedom than the separately excited motor. The separately excited motor is therefore the more suitable choice for this project.

4 Best Motor Configuration for the Load Type

The best motor configuration for the load type means that, in addition to choosing the motor type, the way the field and armature are connected must also match the load. In DC motors, this connection can be series, parallel (shunt) or independent, and each behaves differently in terms of starting torque and speed control. In this project, the selected load is

$$T_L = 1.5\,\text{N}\cdot\text{m}. \tag{4.1}$$

This is a medium load. The motor must therefore be able to supply the required torque, but it does not need a very high starting torque as in traction applications or very heavy loads. Moreover, since the aim of the project is speed control, the selected configuration must allow accurate speed control.

4.1 Comparison of Different DC Motor Configurations

4.1.1 Series Connection

In a series DC motor, the field winding and the armature are connected in series to one source. In this case,

$$i_a = i_f. \tag{4.2}$$

This motor has a high starting torque and is suited to heavy loads. For the $1.5\,\mathrm{N}\cdot\mathrm{m}$ load, however, it is not an optimal choice, because its speed is more difficult to control and the field flux changes with the load.

4.1.2 Parallel (Shunt) Connection

In a shunt DC motor, the field winding is in parallel with the armature, and both are supplied by approximately the same voltage. In this case,

$$V_a = V_f. \tag{4.3}$$

The shunt motor is suited to applications that require an approximately constant speed. It is better than the series motor for speed control, but the field and armature circuits are still not fully independent, so its control flexibility is lower than that of the separately excited motor.

For the $1.5\,\mathrm{N}\cdot\mathrm{m}$ load, the shunt motor can be acceptable but is not the best choice, because the aim of this project is not only to maintain an approximately constant speed but also to design a controller and regulate the motor speed.

4.1.3 Separate Excitation (Independent Connection)

In a separately excited DC motor, the field and armature circuits are supplied from two independent sources. In this case, the field voltage and armature voltage are independent of each other:

$$V_a \neq V_f. \tag{4.4}$$

In this project, the field voltage was kept constant,

$$V_f = 12\,\mathrm{V}, \tag{4.5}$$

and the armature voltage is adjusted by the controller:

$$0 \leq V_a \leq 24\,\mathrm{V}. \tag{4.6}$$

The steady-state field current is

$$i_f = \frac{V_f}{r_f} = \frac{12}{1.43} \approx 8.39\,\mathrm{A}. \tag{4.7}$$

The motor torque is also obtained from

$$T_e = L_{AF}\, i_f\, i_a. \tag{4.8}$$

With i_f constant, torque and speed are easily controlled through the armature current and voltage. For this reason, this configuration is the most suitable option for a constant-torque load.

4.1.4 Permanent-Magnet Motor

In this motor, the field is produced by a permanent magnet and there is no field winding. Its structure is simple, but the field flux cannot be adjusted; it is therefore not an ideal option for this project.

4.2 Final Selected Configuration

Table 1 summarises the comparison of the configurations.

Table 1: Comparison of DC motor configurations for the $1.5\,\mathrm{N}\cdot\mathrm{m}$ load.

Motor configuration	Main feature	Suitability for the $1.5\,\mathrm{N}\cdot\mathrm{m}$ load
Series DC motor	High starting torque; field and armature in series	Speed control more difficult; too powerful for this load
Shunt DC motor	Approximately constant speed; field and armature in parallel	Acceptable, but with less control flexibility
Permanent-magnet DC motor	Simple; no field winding	Field flux cannot be controlled
Separately excited DC motor	Independent field and armature; better speed control	Best choice for this project

Considering the load type and the speed-control objective, the final configuration was selected as follows:

- motor type: separately excited DC motor;
- connection type: independent field and armature supply;
- load type: constant-torque load, $T_L = 1.5\,\mathrm{N}\cdot\mathrm{m}$;
- $V_f = 12\,\mathrm{V}, \quad 0 \leq V_a \leq 24\,\mathrm{V}, \quad \omega_{\mathrm{ref}} = 600\,\mathrm{rad/s}$.

In this structure, the field is supplied with a constant voltage and the armature is controlled by a PI controller. The best configuration for the selected load is therefore the separately excited DC motor with independent field and armature supplies, since it is more controllable than the series connection and offers more flexibility in speed control than the shunt and permanent-magnet motors.

5 Motor Parameters

The parameters used for the separately excited DC motor are listed in Table 2.

Table 2: Parameters of the separately excited DC motor.

Parameter	Value
r_a	0.013
L_{AA}	0.01
V_a	24
V_f	12
T_L	1.5
J	0.21
B_m	1.074×10^{-6}
r_f	1.43
L_{FF}	0.167
L_{AF}	0.004

6 Mathematical Modelling of the Motor

The separately excited DC motor consists of three parts: the field circuit, the armature circuit and the mechanical part. The main equations used in the model are as follows.

6.1 Field-Circuit Equation

$$V_f = r_f\, i_f + L_{FF}\, \frac{di_f}{dt}. \tag{6.1}$$

Transfer function of the field circuit:

$$\frac{I_f(s)}{V_f(s)} = \frac{1}{L_{FF}\, s + r_f}. \tag{6.2}$$

Substituting the parameters:

$$\frac{I_f(s)}{V_f(s)} = \frac{1}{0.167\, s + 1.43}. \tag{6.3}$$

In the steady state:

$$i_f = \frac{V_f}{r_f} = \frac{12}{1.43} = 8.39\,\text{A}. \tag{6.4}$$

The field current will therefore be approximately constant and equal to 8.39 A.

6.2 Armature-Circuit Equation

The armature voltage equation is

$$V_a = r_a\, i_a + L_{AA}\, \frac{di_a}{dt} + e_b, \qquad e_b = \omega\, L_{AF}\, i_f. \tag{6.5}$$

The current is therefore modelled in the Laplace domain as

$$I_a(s) = \frac{1}{L_{AA}\, s + r_a}\big(V_a(s) - e_b(s)\big). \tag{6.6}$$

Substituting the parameters:

$$I_a(s) = \frac{1}{0.01\, s + 0.013}\big(V_a(s) - e_b(s)\big). \tag{6.7}$$

6.3 Electromagnetic Torque

The electromagnetic torque of the motor is calculated from

$$T_e = L_{AF}\, i_f\, i_a. \tag{6.8}$$

Since the field current is approximately constant, an effective motor constant can be defined:

$$K = L_{AF}\, i_f = 0.004 \times 8.39 \approx 0.03356. \tag{6.9}$$

Therefore,

$$T_e = K\, i_a. \tag{6.10}$$

6.4 Mechanical Equation of the Motor

The dynamic equation of the mechanical part of the motor is

$$T_e = J\,\frac{d\omega}{dt} + B_m\,\omega + T_L, \qquad \frac{d\omega}{dt} = \frac{T_e - B_m\,\omega - T_L}{J}. \tag{6.11}$$

In the Laplace domain:

$$\omega(s) = \frac{1}{J\,s + B_m}\big(T_e(s) - T_L(s)\big). \tag{6.12}$$

Substituting the values:

$$\omega(s) = \frac{1}{0.21\,s + 1.074 \times 10^{-6}}\big(T_e(s) - T_L(s)\big). \tag{6.13}$$

7 Motor Characteristic Curve

To examine the motor characteristic, the relation between speed and torque in the steady state is considered. With a constant field current, the value of K is approximately constant:

$$K = L_{AF}\, i_f \approx 0.03356. \tag{7.1}$$

In the steady state, the armature voltage equation is

$$V_a = r_a\, i_a + K\omega, \tag{7.2}$$

and the torque relation is

$$T_e = K\, i_a. \tag{7.3}$$

Hence

$$i_a = \frac{T_e}{K}. \tag{7.4}$$

Substituting,

$$V_a = r_a\,\frac{T_e}{K} + K\omega, \tag{7.5}$$

and therefore

$$T_e = \frac{K}{r_a}\big(V_a - K\omega\big). \tag{7.6}$$

This relation shows that, at constant armature voltage, the torque–speed curve is approximately linear. At $V_a = 24\,\mathrm{V}$, the approximate starting (stall) torque is

$$T_{\text{stall}} = \frac{K}{r_a}\,V_a = \frac{0.03356}{0.013} \times 24 \approx 61.96\,\mathrm{N\cdot m}. \tag{7.7}$$

The approximate no-load speed is

$$\omega_0 = \frac{V_a}{K} = \frac{24}{0.03356} \approx 715.0\,\text{rad/s}. \tag{7.8}$$

At the selected load, $T_L = 1.5\,\text{N}\cdot\text{m}$, the steady-state speed obtained from the analytical relation is approximately

$$\omega_{ss} \approx 697.7\,\text{rad/s}. \tag{7.9}$$

In the simulation, the open-loop steady-state speed was found to be 697.7 rad/s. The equality of these two values shows that the simulation model agrees with the theoretical analysis.

8 Modelling in Simulink

8.1 Open-Loop Model

In the open-loop model, the motor inputs are constant:

$$V_a = 24\,\text{V}, \qquad V_f = 12\,\text{V}, \qquad T_L = 1.5\,\text{N}\cdot\text{m}. \tag{8.1}$$

In this model, the armature voltage is fed directly into the armature circuit. The back-EMF feedback path is taken from the speed output and subtracted from the armature voltage. The field circuit is implemented as an independent transfer function (Figure 1).

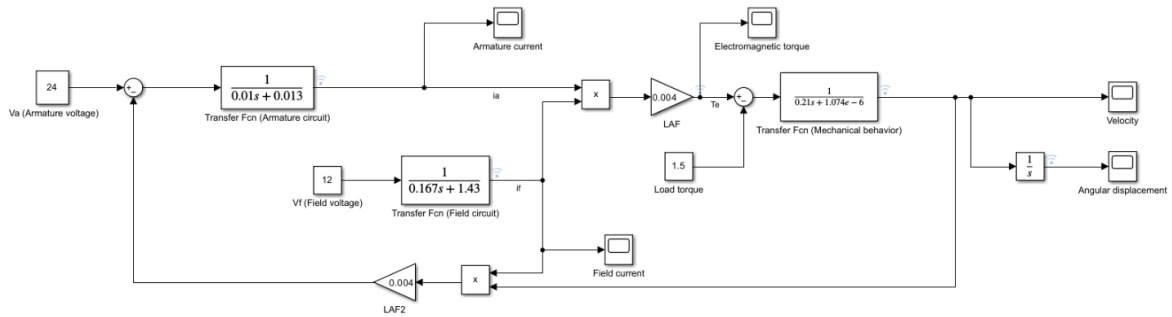


Figure 1: Open-loop model of the separately excited DC motor under a constant-torque load.

8.2 Closed-Loop Model

In the closed-loop model, the constant armature voltage is removed and replaced by the output of the PI controller (Figure 2). The overall structure of the closed loop is

$$\omega_{\text{ref}} \to e(t) \to \text{PI} \to \text{Saturation} \to V_a \to \text{Motor} \to \omega,$$

with the speed feedback

$$\omega \to -\,e(t).$$

The speed error is

$$e(t) = \omega_{\text{ref}} - \omega(t). \tag{8.2}$$

The PI controller generates the appropriate armature voltage, and its output is limited by a Saturation block:

$$0 \le V_a \le 24\,\text{V}. \tag{8.3}$$

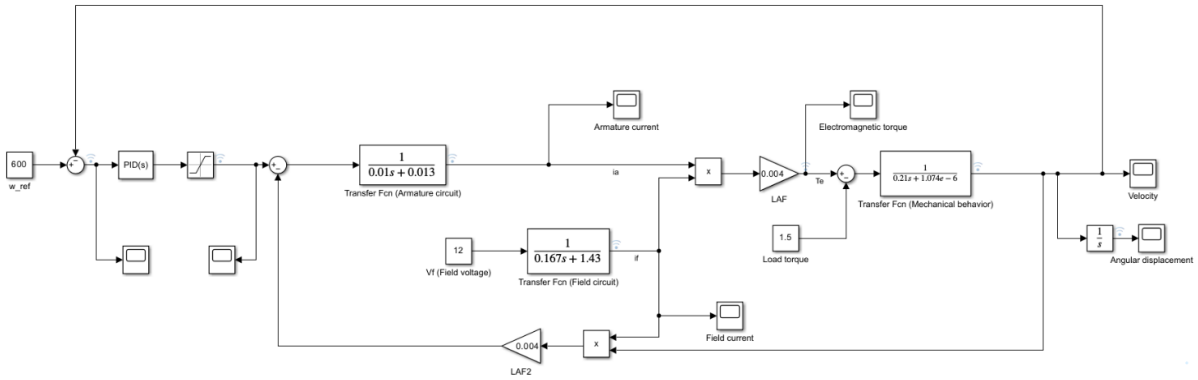


Figure 2: Closed-loop model of the separately excited DC motor with a PI controller.

9 Open-Loop Simulation Results

The open-loop simulation was carried out under the following conditions:

$$V_a = 24\,\mathrm{V}, \quad V_f = 12\,\mathrm{V}, \quad T_L = 1.5\,\mathrm{N\cdot m}, \quad t_{\mathrm{sim}} = 20\,\mathrm{s}. \tag{9.1}$$

9.1 Open-Loop Speed Response

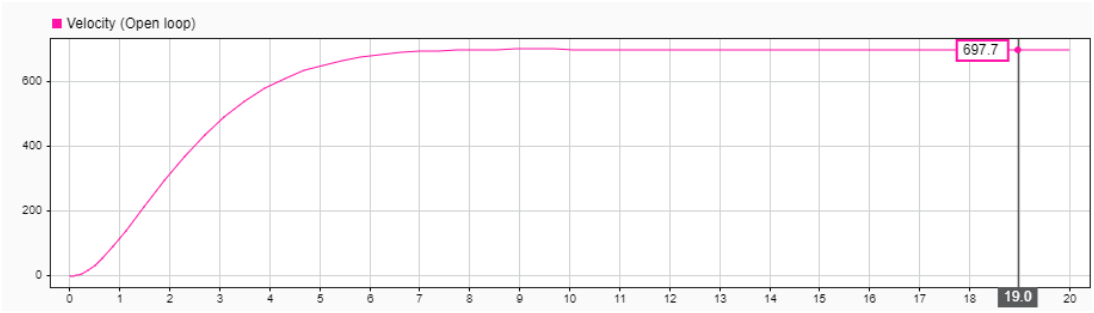


Figure 3: Motor speed response in open loop.

In open loop, the motor speed started from zero and, after the initial transient, reached the following value (Figure 3):

$$\omega_{ss} \approx 697.7\,\mathrm{rad/s} = 6663\,\mathrm{rpm}. \tag{9.2}$$

This result shows that, without a controller, the motor speed is determined solely by the armature voltage, the field voltage and the load, and the speed cannot be set accurately to a desired value.

9.2 Field-Current Response

The field current started from zero and, after a short transient, reached an approximately constant value (Figure 4):

$$i_f \approx 8.4\,\mathrm{A}. \tag{9.3}$$

This value agrees with the calculated value

$$i_f = \frac{12}{1.43} = 8.39\,\mathrm{A}. \tag{9.4}$$

The field circuit has therefore been implemented correctly.

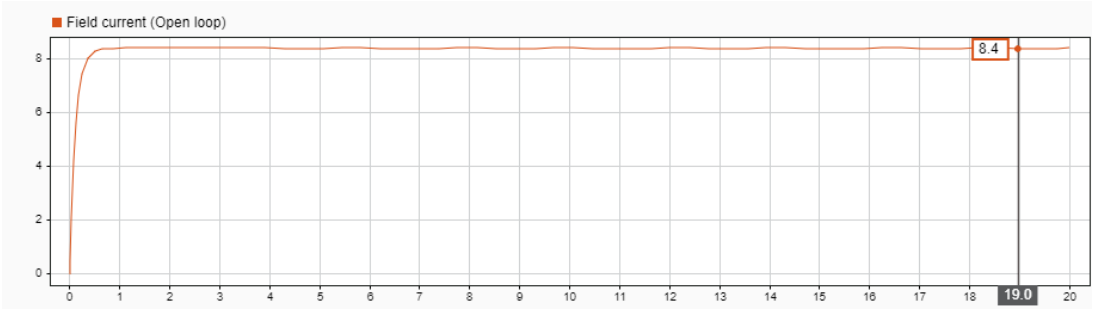


Figure 4: Field-current response of the motor in open loop.

9.3 Armature-Current Response

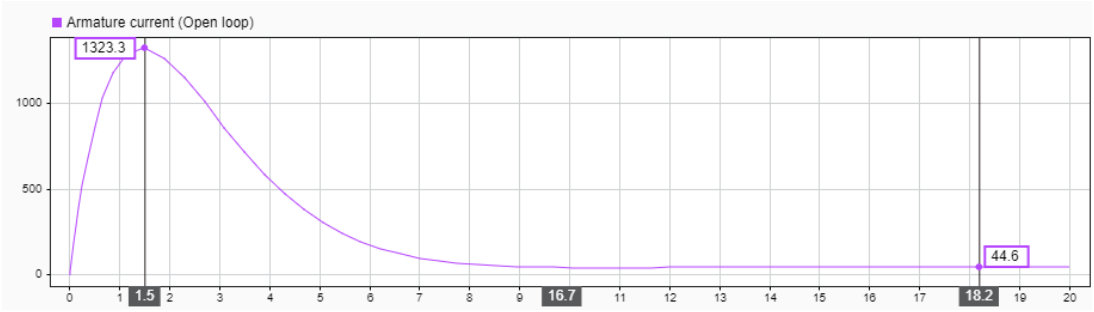


Figure 5: Armature-current response of the motor in open loop.

The armature current is large at the start (Figure 5). This is because, at the instant of starting, the motor speed is zero and there is no back EMF, so the initial current rises. As the speed increases, the back EMF grows and the armature current decreases. In the steady state, since the torque must be approximately equal to the load torque,

$$T_e \approx T_L, \tag{9.5}$$

it follows that

$$i_a \approx \frac{T_L}{K} \approx \frac{1.5}{0.03356} \approx 44.7\,\text{A}. \tag{9.6}$$

The simulation results also show that, after the initial transient, the armature current falls to a low value close to the theoretical one.

9.4 Electromagnetic-Torque Response

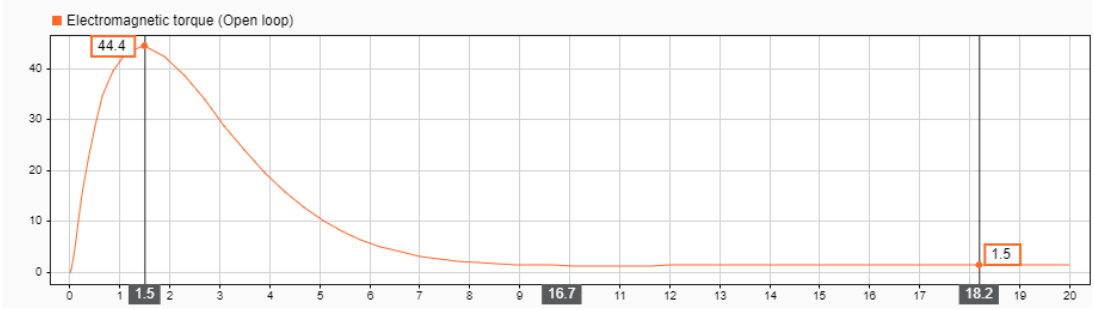


Figure 6: Electromagnetic-torque response of the motor in open loop.

The electromagnetic torque is large at the beginning of the motion, because the motor needs a large torque to accelerate (Figure 6). As the speed increases, the armature current decreases and the torque decreases with it. In the steady state,

$$T_e = T_L + B_m\,\omega. \tag{9.7}$$

Since B_m is very small,

$$T_e \approx T_L, \qquad T_e \approx 1.5\,\mathrm{N}\cdot\mathrm{m}. \tag{9.8}$$

The final torque is therefore equal to the load torque.

9.5 Angular-Displacement Response

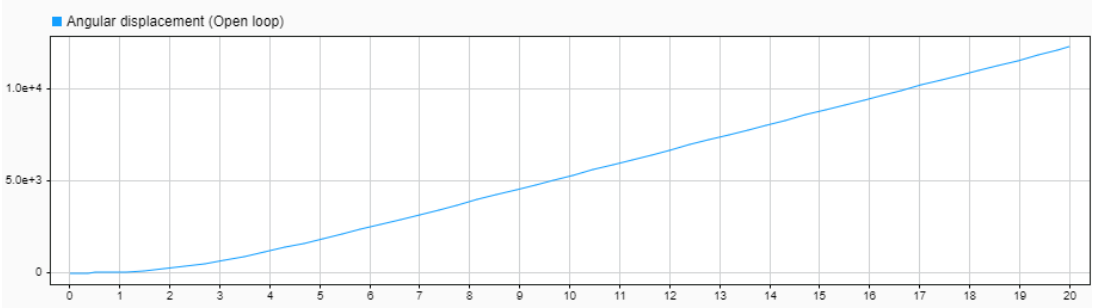


Figure 7: Angular-displacement response of the motor in open loop.

The angular displacement is obtained by integrating the speed:

$$\theta(t) = \int \omega(t)\,dt. \tag{9.9}$$

Once the speed becomes constant in the steady state, the angular displacement increases with an approximately constant slope. This behaviour was observed in the simulation plot (Figure 7) and is as expected.

10 Validation of the Open-Loop Model

To validate the model, the steady-state speed of the motor was calculated analytically.

- Field current:

$$i_f = \frac{V_f}{r_f} = \frac{12}{1.43} = 8.39\,\mathrm{A}. \tag{10.1}$$

- Effective motor constant:

$$K = L_{AF}\,i_f = 0.004 \times 8.39 = 0.03356. \tag{10.2}$$

- In the steady state:

$$i_a = \frac{V_a - K\omega}{r_a}, \qquad T_e = K\,i_a, \qquad T_e = T_L + B_m\,\omega, \tag{10.3}$$

so that

$$K\,\frac{V_a - K\omega}{r_a} = T_L + B_m\,\omega. \tag{10.4}$$

Solving this relation gives

$$\omega_{ss,\text{theory}} \approx 697.7\,\text{rad/s}. \tag{10.5}$$

Simulation result:

$$\omega_{ss,\text{sim}} \approx 697.7\,\text{rad/s}. \tag{10.6}$$

The analytical and simulation results are exactly the same. The open-loop model is therefore mathematically valid.

11 Speed-Controller Design

The aim of the controller is to bring the motor speed to the reference value

$$\omega_{\text{ref}} = 600\,\text{rad/s}. \tag{11.1}$$

This value was chosen because it is lower than the open-loop speed of the motor:

$$600\,\text{rad/s} < 697.7\,\text{rad/s}. \tag{11.2}$$

The motor can therefore reach this speed with a voltage below 24 V. If a reference speed higher than the speed achievable with 24 V had been chosen, the controller would have saturated and the motor could not have reached the desired speed.

11.1 Proportional Controller

A proportional controller was tested first:

$$C(s) = K_p, \qquad K_p = 0.03. \tag{11.3}$$

In this case, the output of the Saturation block no longer locked at 24 V, but the motor speed only reached about 274 rad/s (Figures 8 and 9). The steady-state error was therefore large:

$$e_{ss} = 600 - 274 = 326\,\text{rad/s}. \tag{11.4}$$

The P controller alone is therefore not sufficient for this system.

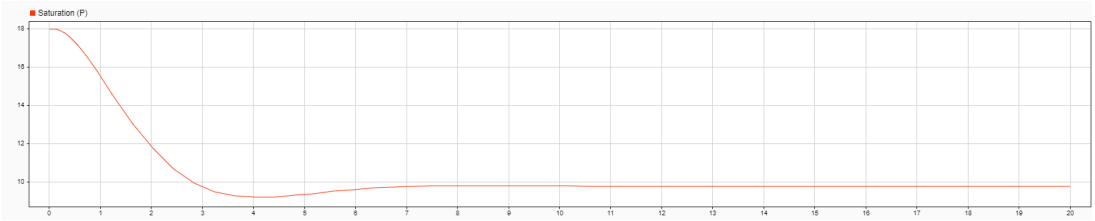


Figure 8: Output of the Saturation block with the P controller.

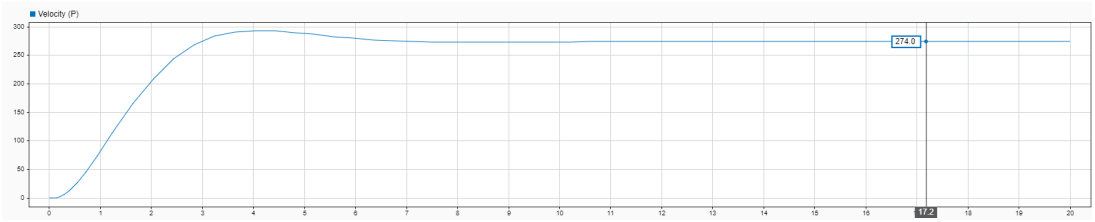


Figure 9: Motor speed response with the P controller.

11.2 PI Controller

To eliminate the steady-state error, an integral term was added to the controller. The final controller is

$$C(s) = K_p + \frac{K_i}{s}, \qquad K_p = 0.035, \quad K_i = 0.015. \tag{11.5}$$

No derivative term was used in this project, because the response of the system with the PI controller was acceptable, and adding a derivative term could have increased the sensitivity of the system to noise (Figures 10 and 11).

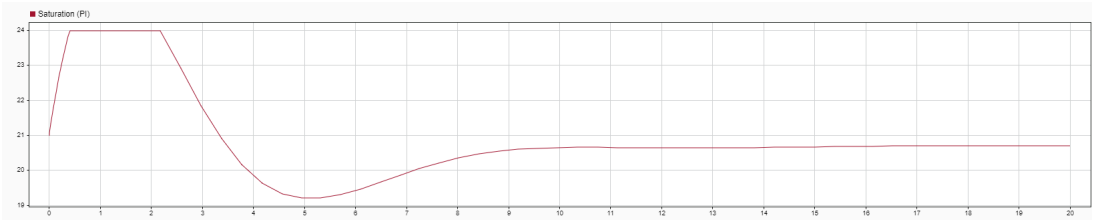


Figure 10: Output of the Saturation block with the PI controller.

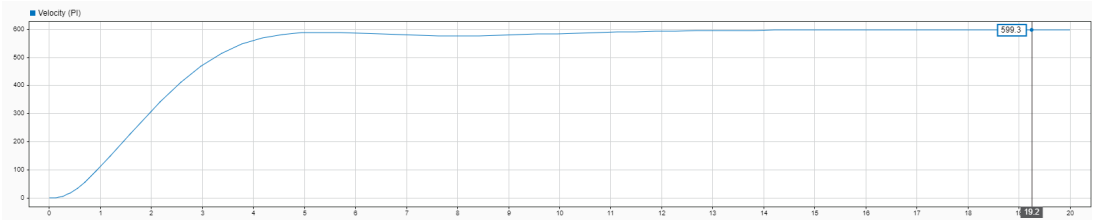


Figure 11: Motor speed response with the PI controller.

12 Closed-Loop Simulation Results

The closed-loop simulation was carried out under the following conditions:

$$\omega_{\text{ref}} = 600\,\text{rad/s}, \quad K_p = 0.035, \quad K_i = 0.015, \quad 0 \le V_a \le 24\,\text{V}, \quad t_{\text{sim}} = 20\,\text{s}. \tag{12.1}$$

12.1 Closed-Loop Speed Response

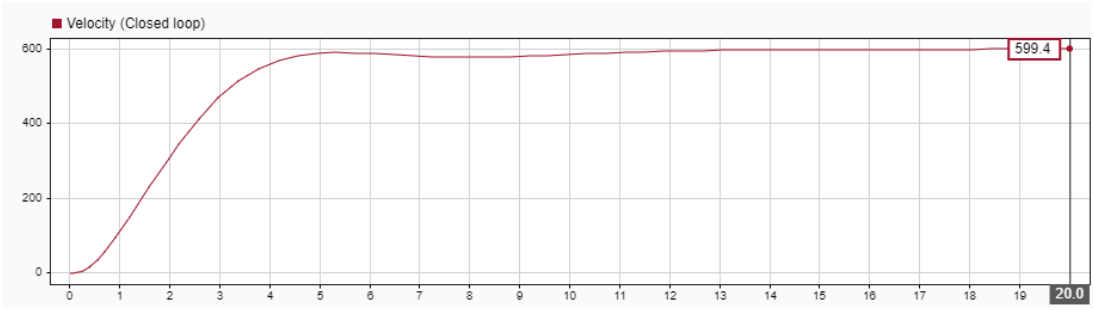


Figure 12: Motor speed response in closed loop.

In closed loop, the motor speed started from zero and, after the initial transient, approached the reference value (Figure 12):

$$\omega_{ss} \approx 599.4\,\text{rad/s}. \tag{12.2}$$

The results show that the PI controller brought the motor speed from its natural open-loop value of about 697.7 rad/s to a value very close to 600 rad/s.

The speed response has a small, acceptable overshoot. The speed first approaches the reference value, oscillates slightly, and then settles on the reference. The steady-state error is almost zero.

12.2 Controller Output and Armature Voltage

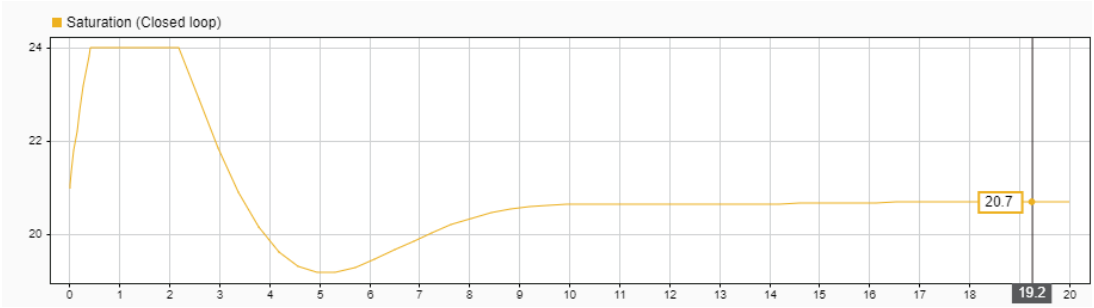


Figure 13: Armature voltage in closed loop.

The controller output is the voltage applied to the armature (Figure 13). At the beginning of the motion, the controller raises the voltage to accelerate the motor, and the output reaches the upper limit of 24 V for a short time. Then, as the speed approaches the reference value, the voltage decreases and finally settles approximately at

$$V_{a,ss} \approx 20.7\,\text{V}. \tag{12.3}$$

This value can also be checked theoretically. In the steady state,

$$V_a = r_a\, i_a + K\omega. \tag{12.4}$$

The steady-state armature current is

$$i_a \approx \frac{T_L}{K} = \frac{1.5}{0.03356} = 44.7\,\text{A}. \tag{12.5}$$

Hence

$$V_a = 0.013\,(44.7) + 0.03356\,(600) = 20.72\,\text{V}. \tag{12.6}$$

This value agrees with the simulation, which gives about 20.7 V.

12.3 Armature Current in Closed Loop

The armature current is high at start-up. This is due to the low armature resistance and the absence of back EMF at the initial instant. As the motor speed increases, the back EMF increases and the armature current decreases (Figure 14).

In the steady state, the armature current reaches a value sufficient to produce the load torque:

$$i_a \approx 46.3\,\text{A}. \tag{12.7}$$

In the simulation plot, too, the armature current decreases after the initial transient and approaches its steady value. The starting current is, however, high. In a real system, a current limiter, a soft starter or a current-control loop is usually used to prevent a large inrush current.

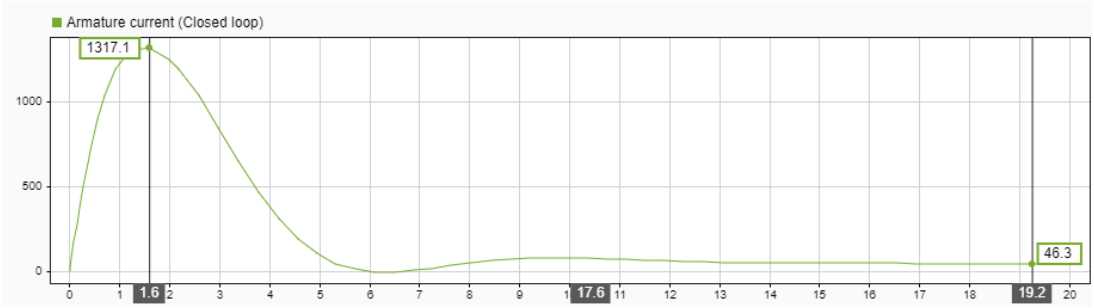


Figure 14: Armature-current response in closed loop.

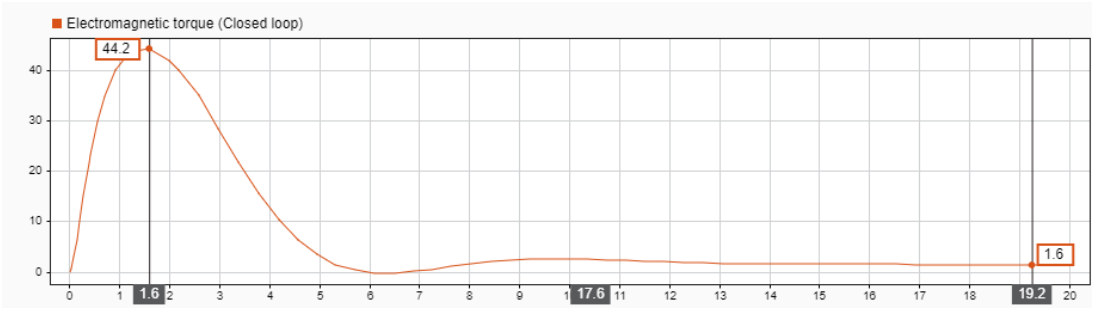


Figure 15: Electromagnetic-torque response in closed loop.

12.4 Electromagnetic Torque in Closed Loop

The electromagnetic torque rises at the beginning of the motion, because the motor needs an initial acceleration to reach the reference speed (Figure 15). After that, as the armature current decreases, the torque also decreases. In the steady state, the motor torque becomes approximately equal to the load torque:

$$T_e = 1.6\,\mathrm{N}\cdot\mathrm{m}, \qquad T_e \approx T_L. \tag{12.8}$$

This result shows that, once the desired speed is reached, the motor produces, to a good approximation, only as much torque as is needed to overcome the load and friction.

12.5 Field Current in Closed Loop

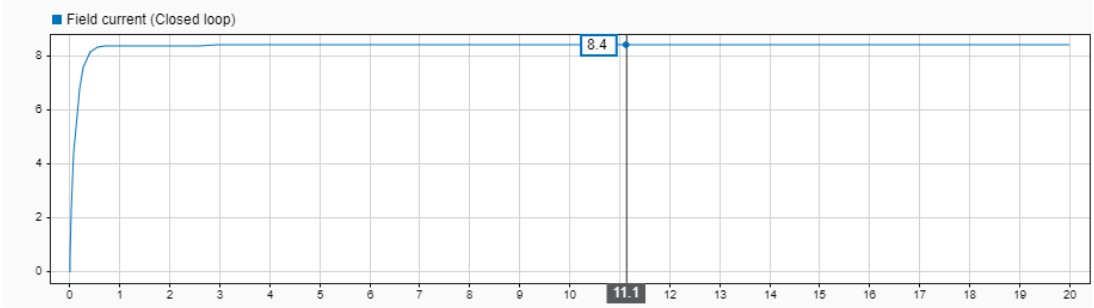


Figure 16: Field-current response in closed loop.

The field current in closed loop is almost the same as in open loop, because the field circuit is independent of the armature circuit (Figure 16). The field voltage is constant at 12 V, so after a short transient the field current reaches the constant value

$$i_f \approx 8.39\,\mathrm{A}. \tag{12.9}$$

This shows that the field flux is approximately constant and that speed control is achieved through the armature voltage.

12.6 Angular Displacement

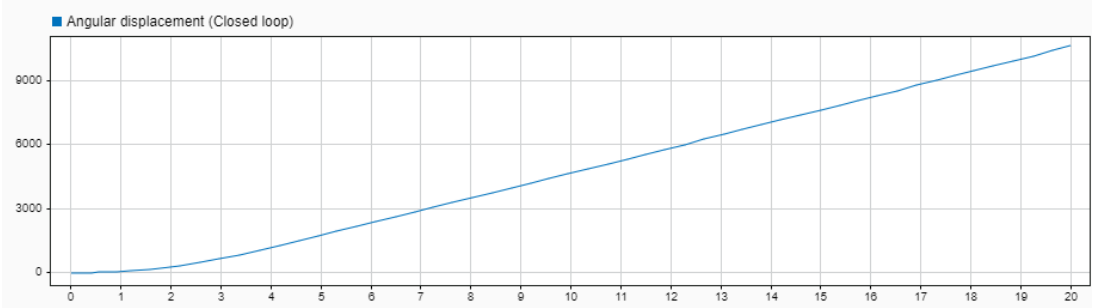


Figure 17: Angular-displacement response in closed loop.

In closed loop, the angular displacement is obtained by integrating the speed (Figure 17). Since the speed finally settles at 600 rad/s, the angular displacement increases approximately linearly after the transient. The slope of this line is smaller in closed loop than in open loop, because the closed-loop steady-state speed is close to 600 rad/s, whereas in open loop it was about 697.7 rad/s.

13 Conclusion

In this project, a separately excited DC motor under a constant-torque load of $T_L = 1.5\,\mathrm{N \cdot m}$ was modelled, simulated and controlled. The main aim of the project was not merely to simulate a DC motor, but to select the motor and a suitable configuration based on the load type, to examine the open-loop behaviour, to design a controller and to analyse the closed-loop performance.

The selected load is a medium constant-torque load. It does not require a very high starting torque, so a series DC motor is not the best choice for it. Because its field and armature are connected in series, the series motor has a high starting torque and is better suited to heavy loads; for the $1.5\,\mathrm{N \cdot m}$ load, however, it is excessively powerful and harder to control in speed. The shunt motor can also be acceptable for medium loads, but since its field and armature are supplied in parallel from one source, its control flexibility is lower than that of the separately excited motor. The permanent-magnet motor is simple, but it does not allow the field flux to be adjusted.

Based on this comparison, the separately excited DC motor with independent field and armature supplies was selected as the best configuration for this project. In this structure, the field and armature circuits are independent. The field voltage was therefore kept constant at 12 V, and the armature voltage was adjusted by the PI controller. With a constant field voltage, the field current reached approximately 8.39 A and the motor flux remained approximately constant. Consequently, the motor torque was controlled mainly through the armature current, which is an appropriate choice for a constant-torque load.

In open loop, the motor was simulated with an armature voltage of 24 V, a field voltage of 12 V and a load of $1.5\,\mathrm{N \cdot m}$. The steady-state speed obtained in the simulation was approximately 697.7 rad/s, and the analytical calculation of the steady-state speed also gave approximately 697.7 rad/s. The closeness of these two values shows that the Simulink model is mathematically valid.

For speed control, a reference of 600 rad/s was chosen, because this value is lower than the speed achievable by the motor in open loop, and the motor can reach it with a voltage below 24 V. A P controller was examined first, but owing to its large steady-state error it did not perform adequately. A PI controller with gains $K_p = 0.035$ and $K_i = 0.015$ was then designed.

The closed-loop results showed that the PI controller brought the motor speed to 599.4 rad/s, i.e. very close to the reference of 600 rad/s. The steady-state controller output, i.e. the armature voltage, was about 20.7 V, which also agrees with the theoretically calculated value. In addition, the steady-state electromagnetic torque became approximately equal to the load torque, and the field current remained constant owing to the independent supply.

In summary, for the $1.5\,\mathrm{N}\cdot\mathrm{m}$ constant-torque load, the separately excited DC motor with independent field and armature connection is the best option among the configurations examined. It is better suited to this load than the series connection, more controllable than the shunt connection, and allows more complete analysis and control than the permanent-magnet motor. The PI controller, with a simple structure, also controlled the motor speed stably and accurately. However, because of the high starting current in the armature circuit, the use of a current limiter or a current-control loop is recommended in practical applications to protect the motor and the drive.

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